



Embodied Intelligence

Brain is the next frontier

Equities
Industrials

Asia

- ◆ Leaders: UniTree (Hardware); X Square/Robbyant (Software); Sharpa/Wuji (Hand); UBTech/Agibot/Deep (Commercialisation)
- ◆ Bottlenecks to Brain's ChatGPT moment: data, data, data; World Model supplements VLA to increase generalisation
- ◆ We prefer hardware less, given intensified price competition; further commercialisation depends on brain breakthrough

We attended the World Robotic Conference and visited various humanoid companies during in 3Q26.

Identifying leaders: Humanoid robots have seen initial signs of commercialisation (Exhibit 10). We now prefer hardware less, given intensified price competition (Exhibit 23, 24). Catalysts include a meaningful breakthrough in brain capability and Optimus 3 commercialisation. Industry consensus leaders along the supply chain are below:

Hardware: UniTree. Its humanoid robots exhibit strong reliability; it has good cost management and it is often used as a development platform for peers.

Brain (software): X Square and Robbyant (Alibaba's investment in humanoid intelligence) are leading in end-to-end world action models.

Dexterous hands: Tier 1-Sharpa (Hesai Technology's founders) and Wuji; Tier 2 players include Brainco, Linkerbot and Inspire Robot.

Commercialisation: Unitree, UBTech and Agibot. Agibot is more balanced in software and hardware. Deep Robotics has a strong customer relationship with State Grid and Southern Grid for power-inspection robots. Dobot has the world's largest collaborative robot shipments, a potential customer base for their humanoids.

Evolving model structure: Lack of physical awareness hinders the wider application of humanoid robots. Understanding of gravity, friction, action and reaction makes humans Newton, and humanoids more human-like.

VLA vs. WM: Previously, the Vision Language Action (VLA) model was dominant in humanoid robot training. World Model is being added to VLA, allowing humanoids to better understand the physical world. This enables humanoids to perform longer-context tasks and allow better generalisation (i.e. less post-training when objects/environments change) versus pure VLA models (Ex. 6). However, we think the definitive path for humanoid models is not final yet, as different startups are trying various structures to improve model performance.

Data, data, data: With open-source data, the key edge for Large Language Models (LLM) is GPUs, a clear US advantage. Humanoid training models have an edge in data collection, a labour-intensive task, which gives Chinese companies an opportunity. **Data recipe:** We see three tiers of data, ranked by cost and usability, namely: 1) real-world data (including teleoperation and UMI data, ex. 5); 2) synthetic data; and 3) egocentric video data. Chinese companies typically use more real-world data, while US players collect more egocentric and synthetic data on cost concerns.

This is our latest report on the Automation theme. If you want to subscribe to any of our nine big themes, [click here](#).

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HSBC Funding the Future Survey

AI megadeals and the rest

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HSBC humanoid robot series reports:

- ◆ [*Embodied Intelligence - UniTree and UBTech: Two different paths for humanoid robots*](#), 14 Apr 2026
- ◆ [*Embodied Intelligence - Humanoid robots \(9\): Dexterous hands are the next focus*](#), 17 Mar 2026
- ◆ [*Humanoid Robots \(8\): Demand driven by government-supported data collection sites*](#), 16 Sep, 2025
- ◆ [*Humanoid Robots \(7\) - Beneficiaries, Catalysts and Reality Check*](#), 8 Apr 2025
- ◆ [*Humanoid Robots \(6\) - Who supplies to Tesla, Unitree and Figure AI?*](#), 28 Feb 2025
- ◆ [*Humanoid Robots \(5\) - Is mass production coming in 2025?*](#), 25 Jan 2025
- ◆ [*Humanoid Robots \(4\) - New arena opens door to new players*](#), 23 Feb 2024
- ◆ [*Humanoid Robots \(3\) - Reviewing progress on Tesla's humanoid robot*](#), 10 July 2023
- ◆ [*Humanoid Robots \(2\): Opportunities, obstacles + the long-awaited Optimus*](#), 8 Sep 2022
- ◆ [*Humanoid Robots: Too early...but opportunities along the value chain*](#), 13 July 2022

Company mentioned on front page

Company	Ticker	Rating	CMP
UBTech	9880 HK	Buy	HKD77.95
Unitree	688836 CH	NR	CNY477.12
Agibot	Private		
Deep	Private		
Robbyant	Private		
X Square	Private		
Sharpa	Private		
Wuji	Private		
BrainCo	Private		
Linkerbot	Private		
Inspire	Private		
Hesai Technologies	2525 HK	NR	HKD16.85

Priced as of 11 Sep 2026

Source: Refinitiv Eikon, HSBC

Peer comparison tables

Exhibit 1: Major humanoid companies comparison – China

Company	UBtech	UniTree	Galbot	Galaxea	Deep Robotics	Dobot	Agibot	X Square	Robbyant
Ticker	9880 HK	688836 CH	Private	Private	Private	2432 HK	Private	Private	Private
Revenue (RMBm)	1,269*	1,152*	na	Na	337	316*	1,050	na	na
Market value** (RMBbn)	37	217	21	20	139	9	40	20	na
Founder	Founder previously worked at Michael Weinig AG	Founder previously worked at DJI	Founder is also the founder of Beijing University EPIC Lab	Founder previously worked at Waymo	Founder previously worked at Zhejinag University	Founder was PhD at Tsinghua University	Founder previously worked at Huawei	Founder previously conducted research in US and was one of the firsts who brought up Attention mechanism	Subsidiary of Ant Group
R&D expense (RMBm)	303*	136*	na	na	84	102*	na	na	na
Focus industry	Manufacturing, To-C	Educational, Commercial	Service, To-C	Service	Security, inspection	Manufacturing, service	Commercial	To-C	Software

*1H26

**Publicly reported valuation for private companies

Source: Company data

Exhibit 2: Major humanoid companies comparison – Overseas

Company	Tesla	Physical Intelligence	Generalist AI	Skild AI	Field AI	Figure AI	Agility Robotics	1X
Ticker	TSLA US	Private	Private	Private	Private	Private	AGLT US*	Private
Revenue (USDm)	94,827	na	na	na	na	na	na	na
Market value** (USDbn)	1,454	na	2	14	2 (in 2025)	na	2.5	6-10
Founder	Elon Musk is a key person	Founder was the co-founder of 1X	Founder previously worked at DeepMind	Founders were professors at Carnegie Mellon University	Founder previously worked at NASA	Founder also founded Vetterly and Arhcer Aviation	Founders were PhDs at Carnegie Mellon University	1X was founded in 2014 under the original name Halodi Robotics
R&D expense (USDm)	6,411	na	na	na	na	na	37	na
Focus industry	Manufacturing	General purpose (Software)	General purpose (software)	General purpose (software)	Construction	To-C	Manufacturing	To-C

*Agility to merge with Churchill Capital XI (CCXI), expected to close in 4Q 2026.

**Publicly reported valuation for private companies

Source: Company data

Exhibit 3: Major dexterous hand companies comparison

Company	Sharpa	Paxini	Linkerbot	Wuji	BrainCo	Inspire
Revenue (RMBm)	Expected to generate revenue from 3Q26	na	260	na	na	na
Market value (RMBbn)	22	10	42	na	na	na
Founder	Founding team also owns Hesai (2525 HK)	Founder previously conducted research at Waseda University	Founder has more than 10 years of experiences in robotics.	Founder graduated from UIUC.	Founder was PhD at Harvard University	Founder has 20 years of experiences in robotics

Source: Company data

Leaders along the supply chain

Exhibit 4: Industry consensus leaders in each area

Area	China leaders	US leaders
Hardware	UniTree	Boston Dynamics*, Agility Robotics
Brain	X Square, Robbyant	Physical Intelligence, Figure AI
Dexterous hands	Sharpa, Wuji	Tesla, Figure AI
Commercialisation	UBTech, Agibot, Deep Robotics	Figure AI, Agility Robotics
Beta + Alpha	Dobot (Cobot + Humanoid)	Tesla (EV + Humanoid)

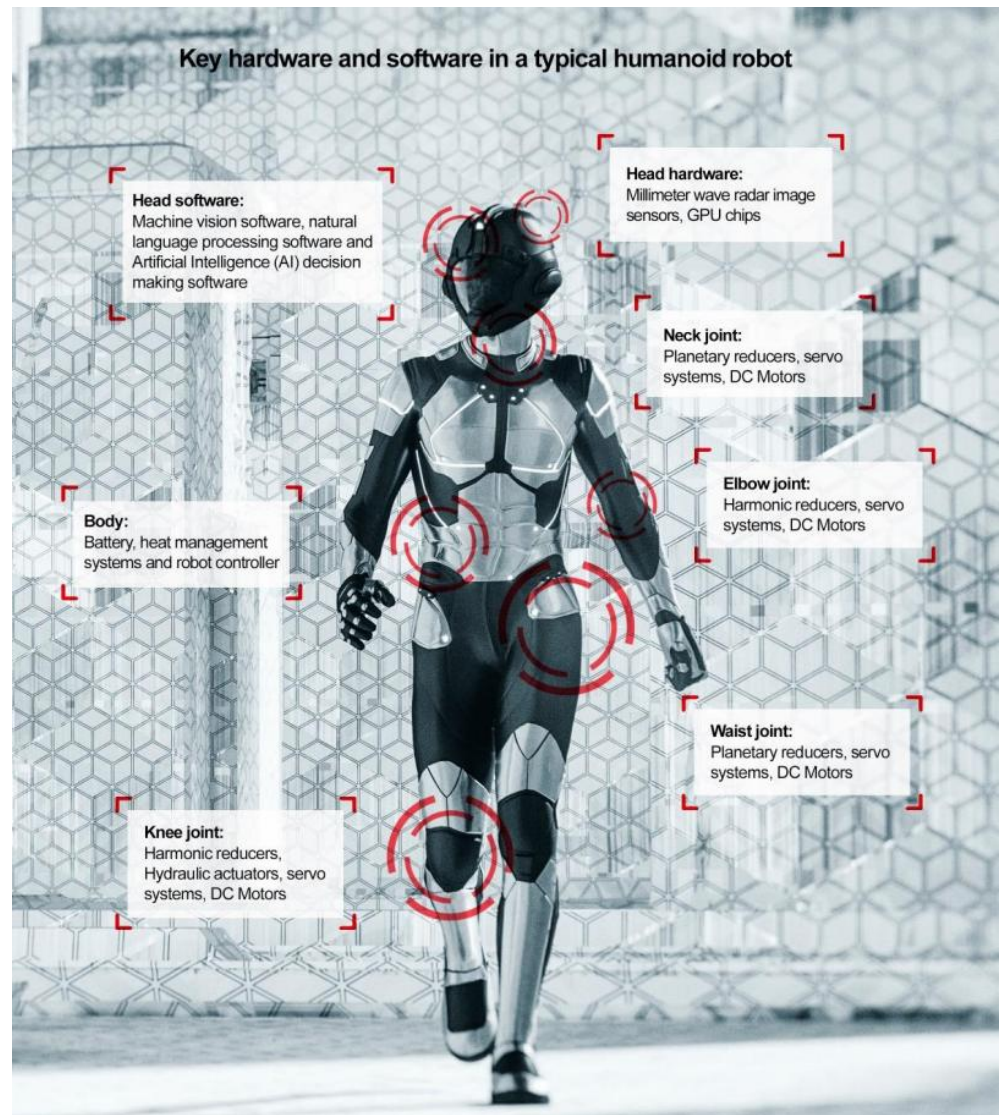
*Acquired by Hyundai Motor

Source: Humanoid industry experts opinions

Exhibit 5: HSBC humanoid robot market size estimates scenario analysis

	2025	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	CAGR 2025-30e	CAGR 2025-34e
Total units (k units)												
Bear case	13	20	39	79	236	472	707	920	1,150	1,379	105%	68%
Base case	16	31	62	187	562	1,123	1,685	2,359	3,066	3,680	135%	83%
Bull case	18	91	362	1,086	3,258	6,516	9,774	12,706	15,883	19,059	225%	117%
Market size (USDbn)												
Bear case	0.8	1.0	1.7	2.9	6.9	12.4	17.6	21.7	26.4	30.7	75%	51%
Base case	0.9	1.4	2.3	5.6	14.2	25.5	35.2	46.8	60.9	73.1	95%	63%
Bull case	1.0	3.7	11.0	26.5	65.1	130.2	195.2	253.8	317.3	380.7	162%	92%
Average selling price (USDk)												
Bear case	58	52	44	36	29	26	25	24	23	22	-15%	-10%
Base case	58	46	37	30	25	23	21	20	20	20	-17%	-11%
Bull case	58	41	30	24	20	20	20	20	20	20	-19%	-11%

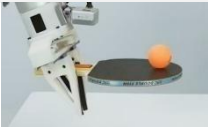
Source: Company data, HSBC estimates

Exhibit 6: Humanoid robot hardware and software components


Source: HSBC

Brain: Training data and model

Exhibit 7: Humanoid training data hierarchy

Data tier	Data type	Illustration	Quality/Cost	Pros	Cons	Cost (USD/hr)	Average usability
1	Real-world on-policy data		High	Most useful	Expensive	90	
1	Real-world teleoperation data		Mid-high	Highly useful	Expensive	45	90%
1	Real-world UMI data	 *	Mid	Useful	Good balance between usability and cost	10	80%
2	Synthetic data		Mid	Fast iteration and scalable	Sim-to-real gap, higher computing requirement		
3	Human behavioural data (Egocentric video)		Mid-low	Lower cost of acquisition	Less useful versus real-world data	9	60%
3	Internet data		Low	Widely available	Usability		

*UMI employs hand-held grippers, which can also be installed on a robot and used teleoperationally, as shown in the illustration

Source: Company data, HSBC

Exhibit 8: VLA model versus WAM

Type of foundation model	Visual Language Action (VLA) Model	World Action Model (WAM)
Popularity	Widely adopted, backbone of robot brain	Newly emerging model that supplements VLA
Compute requirement	>1,000 GPU cards	>10,000 GPU cards
Pros	Fast instruction following	Naturally suited to long-horizon tasks
Cons	Struggles with multi-step tasks	Higher computing cost
Best for	Moving, Pick and place	Long-horizon tasks

Source: Company data, HSBC

Exhibit 9: Figure AI's "Helix" VLA training compared with traditional imitation training

Traditional imitation training			
Training method		Characteristics	
Task 1	• 500 hours of training • Neural model 1	Seperate control for different parts	Single-robot control
Task 2	• 500 hours of training • Neural model 2		
Task 3	• 500 hours of training • Neural model 3	Only performs pre-trained tasks	Model runs on-board or through cloud

Figure AI's "Helix" Vision-Language-Action (VLA) Model			
Training method		Characteristics	
Task 1	• 500 hours of training in total • One single VLA model • New behaviors scale with no programming or demonstrations	Full-upper body control through VLA	Multi-robot collaboration
Task 2		Pick up anything on natural language prompts	VLA runs entirely on GPUs on-board
Task 3			
...			

Source: HSBC compilation

Commercialisation: Shipments, financials & breakthroughs

Exhibit 10: Humanoid robot shipments

	2024	% of total shipment	2025	% of total shipment
UBTech	250	11%	1,079	6%
UniTree	800	35%	5,500	31%
Agibot	600	26%	5,168	29%
Leju	100	4%	500	3%
Engine AI	0	0%	400	2%
Fourier	0	0%	300	2%
Galbot	0	0%	1,200	7%
Other in China	400	17%	2,053	11%
China Total	2,150	93%	16,200	90%
Tesla	50	2%	150	1%
RoW	100	4%	1,650	9%
World Total	2,300	100%	18,000	100%

Source: Company data, Omdia, IDC, HSBC

Exhibit 11: UBTech and UniTree humanoid volume and ASP

	2023	2024	2025	1H26
UBTech				
Sales volume (unit)		10	1,079	921
	y-o-y		10690%	1947%
ASP (RMBk)		3,562	760	641
	y-o-y		-79%	-25%
Unitree				
Sales volume (unit)	5	410	5,500	5,900
	y-o-y	8100%	1241%	
ASP (RMBk)	593	261	158	
	y-o-y	-56.1%	-39.5%	

Source: UniTree Prospectus, UBTech, SAG & Huxiu, HSBC

Exhibit 12: Listed Chinese humanoid robot companies' key financials comparison, 1H26

Company Ticker		UBTech 9880 HK	UniTree 688836 CH	Dobot 2432 HK
Revenue (RMBm)		1,269	1,152	316
	Revenue y-o-y	104.2%	48.5%	106.5%
Gross profit (RMBm)		567	645	150
	GPM	44.7%	56.0%	47.4%
Net income/(loss) (RMBm)		(311)	274	(106)
	NPM	-24.5%	23.8%	-33.4%
R&D expense (RMBm)		(303)	(136)	(102)
	as % of revenue	23.9%	11.8%	42.8%
Market cap (RMBbn)		37	217	9

Source: Company data, HSBC estimates

Exhibit 13: Humanoid robot products comparison

	Height (cm)	Weight (kg)	Range	Launch	Price (RMB)
UBTech					
Walker S1	172	76	Full-size, Industrial	Oct-24	
Walker S2	176	70	Full-size, Industrial	Mid 2025	c600k
Tiangong	172	n/a	Full-size, Research	Mar-25	c300k
UniTree					
H1	180	47	Full-size	Aug-23	
G1	132	35	Mid-size	May-24	From 85k
R1	123	25	Consumer-level	Jul-25	From 29.9k
H2	182	70	Full-size embodied	Oct-25	From 205k
Dobot					
Atom (base)	153	62	Full-size, Industrial/commercial	Mar-25	c200k
Atom Max	165	60	Full-size, Research/Industrial AI R&D	Mid 2025	
Atom Trainer	165	n/a	Full-size, Research/Education	Mid 2025	

Source: Company data

Exhibit 14: China humanoid robot companies and recent breakthrough

Company	Humanoid robot	Brain/Body milestones	Dexterous hand milestones
Unitree Robotics (宇树科技)	H2, G1	Debut of its fully autonomous cluster control technique, quick movement up to 4m/s with no external positioning assistance required	Dex5 has 20 degree of freedom (including 16 active ones and 4 passive ones)
Agibot (智元机器人)	A2, X2, G2, D1	Brought up SOP (Scalable online post-training)	NA
Leju Robotics (乐聚机器人)	Kuabvo	First humanoid to equip 5G-A technology. First humanoid to equip Huawei's HarmonyOS. Hardware self-development ratio is as high as 90%	First in China to open source humanoid robot training data in real world settings
Galbot (银河通用)	G1 (wheeled)	The world's first end-to-end large model called "AstraBrain" that integrates "cerebrum-cerebellum-neural control" in one unit and can control body and hand at the same time. GroceryVLA, a part of AstraBrain, is the world's first end-to-end embodied large model targeting retail.	More than 30 motors and has 22 degree of freedom.
MagicLab (魔法原子)	MagicBot Z1, Gen 1, MagicDog	Hardware self-development ratio as high as 90%	Self-developed MagicHand S01 has 11 degree of freedom, can load up to 5 kg of weight per hand, and is capable of performing over 50 fine actions
Noetix Robotics (松延动力)	Bumi, N2, E1	Bionic face with 32 motion control motors (including 12 for mouth) and multimodal collaborative expression model for facial movement.	NA
Galaxea (星海图)	R1	Official launch of end-to-end VLA G0 Plus and first out-of-box VLA demo in the world called <i>Pick Up Anything</i>	NA
UBtech	Walker, C1, U1	Launched Uworld U1 series, an ultra-bionic humanoid robot with biomimetic silicone skin, 88 degrees of freedom, and a dual-pivot cervical spine.	NA
Deep robotics	DR01, DR02	Na	NA
Dobot	Atom	Na	NA
X Square	Quanta series	Launched global first World Unified Model called WALL.	NA
Robbyant	Na	Launched LingBot 2 world model, including LingBot-Video, a first-in-global open-sourced MoE video generation model for humanoid.	NA

Source: HSBC

Exhibit 15: Orders received by Chinese humanoid robot companies

Company	Time of confirmation	Order size (RMBm)	Client	Category
UBTECH	04/25	50	na	Auto
	07/25	91	MIEE Automotive (觅亿汽车)	Auto
	09/25	250	na	Industrial
	09/25	30	Miracle automation (天奇自动化)	Auto
	10/25	126	Guangxi Liuzhou Embodied Intelligence (柳州具身智能)	Data collection
	10/25	32	na	Auto
	11/25	159	Zigong Shoutou Robot Industry Investment (自贡数投机器人)	Data collection
	11/25	264	Guangxi Fangchenggang (广西防城港)	Data collection
	11/25	143	Jiangxi Jiujiang (江西九江)	Data collection
	12/25	60	Guangdong Huizhou (广东惠州)	Data collection
	12/25	50	Zoellent Technology (卓世科技)	Data collection
	12/25	78	Inner Mongolia Huhehaote (内蒙古呼和浩特)	Data collection
	12/25	na	Texas Instrument	Industrial
	01/26	na	Airbus	Aero
Total		>1,300		
Agibot	06/25	28	na	Industrial
	06/25	25	Zhejiang Shangyu (浙江上虞)	Data collection
	07/25	double digits	Chery automobile	Auto
	07/25	78	China Mobile (中国移动)	Business service
	07/25	13	Guangdong Zhuhai (广东珠海)	Data collection
	08/25	double digits	Fulin Precision (富临精工)	Industrial
	09/25	31	Hubei Humanoid Robotics Innovation Center (湖北人形机器人创新中心)	Data collection
	10/25	over 100	Longcheer (龙旗科技)	3C
	10/25	over 100	Joyson (均胜电子)	Auto
	02/26	na	Mint Group	Auto
Total		>400		
Unitree	03/25	8	Tongji University (同济大学)	Data collection
	07/25	46	China Mobile (中国移动)	Business service
Total		54		
Galbot	12/25	~700	Baida (百达精工)	Industrial
Dobot	10/25	81	Red fairy (瑞德丰精密)	Industrial
	10/25	na	Inspur	Data collection
	11/25	~200	Lens Technology (蓝思科技)	Industrial
Total		~280		
Leju Robotics	08/25	83	SOE	Data collection

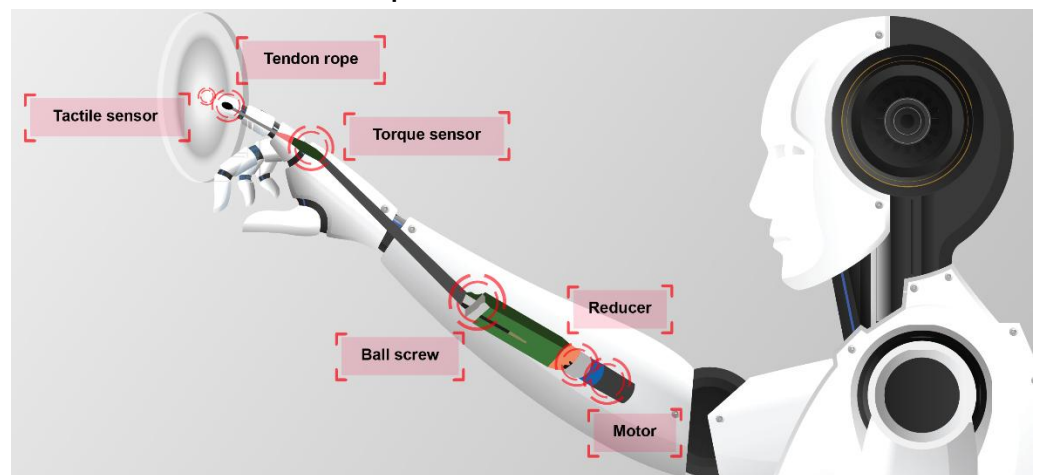
Source: HSBC summarised from public information

Dexterous hands

We believe dexterous hands are in an earlier stage of development than humanoid robots, given two challenges: reliability and model compatibility. Current models of dexterous hands often doesn't meet the reliability requirement in factory production lines. In addition, the high degree of freedom makes model training difficult, often requiring re-training for different tasks and hand configurations.

As a result, we observe that most humanoid robots in factories use grippers instead of dexterous hands. However, the industry consensus view is that Sharpa and Wuji's dexterous hands lead in technology. In terms of hand design, tendon rope mechanism is increasingly being used in high degree of free hands. HSBC estimates global dexterous hand shipment of c20k units in 2025, c70k in 2026.

Exhibit 16: Dexterous hands component illustration



Source: HSBC

Exhibit 17: Tesla's Optimus dexterous hands progression

Tesla	Gen 1	Gen 2	Gen 3
Launch at	2022/10	2023/12	2024/10
Motion control	Coreless motors	Coreless motors	Coreless motors
Reducer	Planetary reducer	Planetary reducer	Planetary reducer
Transmission	Worm gear	Worm gear	Ball screw + Tendon rope
DoF	11 (6 active + 5 passive)	11 (6 active + 5 passive)	22 (17 active + 5 passive)
Number of motors	6	6	17
Sensor	None	Tactile sensors on finger tips	Tactile sensors over entire hand
Functions	Can perform simple catch and release tasks	Can perform precise tasks such as folding shirts	Can perform comprehensively dynamic tasks such as receiving flying tennis ball

Source: HSBC

Exhibit 18: Dexterous Hands' bill of materials (BOM) cost estimates (RMB)

Humanoid robot joints/parts classification			Main suppliers	High-end				Low-end			
				Number	Unit price	Single hand value	% of cost *	Number	Unit price	Single hand value	% of cost
Dexterous Hand	Motion control	Coreless motor	Moons' Electric (鸣志电器), Leili Motor (江苏雷利), Zhaowei Machinery & Electronics (兆威电机), DINGS Intelligent (鼎智科技), Topband (拓邦股份)	10-15	1,000-2500	15,000-37,500	44%	7	300	2,100	39%
	Reducer	Planetary reducer	Zhaowei Machinery & Electronics (兆威电机), ZD motor (中大力德)	10-15	100-200	1,000-3,000	4%	7	200	1,400	26%
	Transmission	Ball screw	XCC Group (五洲新春), WuxiBest (贝斯特), Beite (北特科技), Hengli Hydraulics (恒力液压), Seenpin (新剑传动), Hiwin (上银科技)	~20	300-500	6,000-10,000	13%	7	150	1,050	20%
		Tendon rope	Hanvo Safety (恒辉安防), Nanshan Fashion (南山智尚), Tongyizhong New Material (同益中), Daye Shares (大业股份)	~40	200-300	8,000-12,000	17%				
	Sensor	Tactile sensor	PaXini (帕西尼), Tashan Tec (它山科技), XENSE ROBOTICS (千觉机器人), Matrix (墨现科技)	10-15	500	5,000-7,500	10%	5	100	500	10%
		Torque sensor	Keli (柯力传感), Link Touch (蓝点触控), Sunrise Instruments (宇立仪器)			5,000-10,000	13%	7	50	350	6%
	Whole unit		Inspire Robots (因时机器人), Linkerbot (灵心巧手), Zhaowei Machinery & Electronics (兆威电机), OY Motion (傲意信息科技), HitBot (慧灵科技)								
Total Cost						40,000-80,000	100%	5,400			100%

*at midpoint.

Source: HSBC estimates

Exhibit 19: Domestic dexterous hands comparison – high-end

	Agibot OmniHand Pro	Unitree Dex5-1	Galbot G1 dexterous hand	Linkerbot L30 (Tendon/gear version)	Zhaowei ZWHand DM17/20	Sharpa Wave	Paxini DexH13	Wuji Hand
Launch at	2025/8	2025/4	2026/2 (At China's Spring Festival Gala)	2025	2026/1	2026/4	Na	2025/9
Motion control	Coreless motors	Coreless motors	Coreless motors	Coreless motors	Coreless motors	Coreless motors	Coreless motors	Coreless motors
Reducer	Planetary reducer	Planetary reducer	Harmonic reducer + planetary reducer	Planetary reducer	Planetary reducer	Na	Na	Na
Transmission	Ball screw + rod	Gear + ball screw	Gear	Tendon or Gear	Gear	Gear	Gear	Gear
DoF	19 (12 active + 7 passive)	20 (16 active + 4 passive)	22	25 (18 active + 7 passive) Or 21	17-20	22	16 (13 active + 3 passive)	20
Number of motors	Na	16	Over 30	21 or more	17-20	22	13	20
Sensor	Over 150 tactile sensors	Total of 94 tactile sensors	Tactile sensors/torque sensors over entire hand	Multiple sensors	Tactile sensors on finger tips	Tactile sensors on finger tips	Tactile sensors on finger tips, embedded AI camera	Na
Functions	Can perform precise tasks	Can perform comprehensively dynamic tasks such as playing poker/turning pages	Can perform comprehensively dynamic tasks such as manipulating walnut	Can perform precise tasks	Na	Can perform precise tasks	Can perform precise tasks	Can perform precise tasks

Source: HSBC

Hardware components experience fierce competition

Exhibit 20: Humanoid robot components and related companies

Component	Global suppliers	Mainland China suppliers
Actuator assembly		Zhejiang Sanhua (三花智控), Tuopu (拓普)
Drive and motors	Kollmorgen, Moog, Maxon Motor, BEI Kimco, Woodward, Shinano Kenshi, Nidec, Bosch, TQ Robodrive, Faulhaber, Maxon, Portescap, Allied Motion	Shenzhen Inovance (汇川), Kinco (步科), Moons (鸣志电器), Haozhi (昊志机电), Dings Motion (鼎智科技), HMC (恒帅), Leili (江苏雷利)
Precision reducers	Harmonic Drive, Nabtesco	Leader Drive (绿的), Zhongda Leader (中大力德), Shuanghuan (双环传动)
Ball screws	THK, NSK, Rexroth, Schaeffler, Rollvis, GSA, Ewellix	Beite (北特科技), Hengli (恒立液压), Best (贝斯特)
Bearings	JTEKT, NSK, MinebeaMitsumi	XCC (五洲新春), Lixing (力星股份), Changsheng (长盛轴承), Sinomach Precision (国机精工).
Chips	NVIDIA, Intel, AMD	Huawei (华为), Cambricon (寒武纪)
Image sensors	Intel, Sony, Samsung	Smartway (斯特威), Galaxycore (格科微)
Force sensors	ATI, Robotous, AMTI	XDLK Microsystem (芯动联科) Keli (柯力传感), Sunrise Instruments (宇立仪器)
Millimetre wave radar	Texas Instruments, Bosch, Valeo, Continental, NEC	Desay SV (德赛西威), Hasco (华域汽车)
AI software	Tesla, Nvidia, Figure AI	SenseTime (商汤), Baidu (百度)
Battery cell	Panasonic, LG Energy	CATL (宁德时代)

Source: Company data, HSBC compilation

Exhibit 21: Humanoid robots' bill of materials (BOM) cost estimates (RMB)

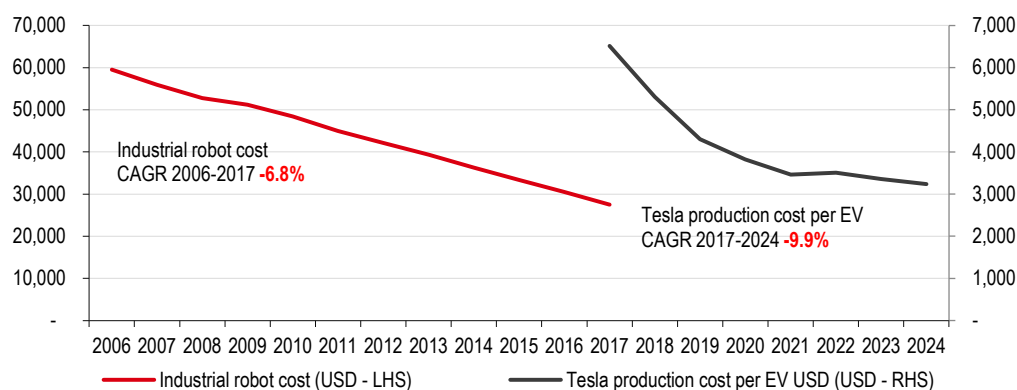
Humanoid robot joints/parts classification			Main suppliers	Current stage cost				Final stage cost			
				Number	Unit price	Single machine value	% of cost	Number	Unit price	Single machine value	% of cost
Joint	Rotary actuator*14	Servo drive	Inovance Technology (汇川科技), Hechuan Technology (禾川科技), Blue Ocean Huateng (蓝海华腾)	1	300	4,200	2.0%	1	100	1,400	1.4%
		Frameless torque motor	Kinco (步科股份), Hechuan Technology (禾川科技), HAOZHI (昊志机电)	1	1,000	14,000	6.8%	1	500	7,000	7.2%
		Harmonic reducer	Leaderdrive (绿的谐波), FORE (丰立智能), Shuanghuan (双环传动)	1	1,000	14,000	6.8%	1	600	8,400	8.7%
		Encoder (position sensors)	UP Optotech (奥普光电), Inovance Technology (汇川科技), Hechuan Technology (禾川科技)	2	400	11,200	5.4%	2	200	5,600	5.8%
		Mechanical clutch		1	80	1,120	0.5%	1	50	700	0.7%
		Crossed roller bearings	XCC (五洲新春), Lixing(力星股份), Changsheng (长盛轴承), Sinomach Precision (国机精工)	1	150	2,100	1.0%	1	100	1,400	1.4%
		Angular contact bearings		2	100	2,800	1.4%	2	50	1,400	1.4%
		Total	Tuopu (拓普集团), Sanhua (三花智控)		3,530	49,420	24.0%		1,850	25,900	26.8%
	Linear actuator*14	Servo drive	Inovance Technology (汇川科技), Hechuan Technology (禾川科技), Blue Ocean Huateng (蓝海华腾)	1	300	4,200	2.0%	1	100	1,400	1.4%
		Frameless torque motor	Kinco (步科股份), Hechuan Technology (禾川科技), HAOZHI (昊志机电)	1	1,200	16,800	8.2%	1	500	7,000	7.2%
		Planetary roller/ ball screw/ Trapezoidal lead screw	Beite Technology (北特科技), Best Precision Machinery (贝斯特), Hengli Hydraulic (恒立液压), XCC (五洲新春)	1	3,000	42,000	20.4%	1	1,000	14,000	14.5%
		Encoder (position sensors)	UP Optotech (奥普光电), Inovance Technology (汇川科技), Hechuan Technology (禾川科技)	1	400	5,600	2.7%	1	200	2,800	2.9%
		Deep groove ball bearings	XCC (五洲新春), Lixing (力星股份), Changsheng (长盛轴承), Sinomach Precision (国机精工)	1	100	1,400	0.7%	1	50	700	0.7%
		Four-point contact ball bearing		1	50	700	0.3%	1	30	420	0.4%
			Total	Tuopu (拓普集团), Sanhua (三花智控)		5,050	70,700	34.4%		1,880	26,320
	Dexterous Hand*2	Coreless motor	Moons' Electric (鸣志电器), Leili Motor (江苏雷利), Zhaowei Machinery & Electronics (兆威电机), DINGS Intelligent (鼎智科技), Topband (拓邦股份)	7	300	4,200	1.0%	7	200	2,800	2.9%
		Planetary reducer	Zhaowei Machinery & Electronics (兆威电机), ZD motor (中大力德)	7	200	2,800	0.7%	7	100	1,400	1.4%
		Ball screw	XCC Group (五洲新春), WuxiBest (贝斯特), Beite (北特科技), Hengli Hydraulics (恒力液压), Seenpin (新剑传动), Hiwin (上银科技)	7	150	2,100	0.5%	7	100	1,400	1.4%
		Tendon rope	Hanvo Safety (恒辉安防), Nanshan Fashion (南山智尚), Tongyizhong New Material (同益中), Daye Shares (大业股份)								0.0%
		Tactile sensor	PaXini (帕西尼), Tashan Tec (它山科技), XENSE ROBOTICS (千觉机器人), Matrix (墨现科技)	5	100	1,000	0.2%	5	50	500	0.5%
		Torque sensor	Keli (柯力传感), Link Touch (蓝点触控), Sunrise Instruments (宇立仪器)	7	50	700	0.2%	7	50	700	0.7%
			Total	Inspire Robots (因时机器人), Linkerbot (灵心巧手), Zhaowei Machinery & Electronics (兆威电机), OY Motion (傲意信息科技)		5,400	10,800	5.3%		3,400	6,800
		Subtotal of Joints				125,520	61.1%		59,020	61.0%	
Battery	Battery Pack	CATL (宁德时代), EVE Energy (亿纬锂能), BYD (比亚迪)	1	5,000	5,000	2.4%	1	2,500	2,500	2.6%	
Tactile Sensor	Camera	OPT (奥普特), Smartway (斯特威), Galaxcore (格科微)	3	500	1,500	0.7%	3	300	900	0.9%	
	IMU	Xdlk (芯动联科), Memsensing Microsystems (敏芯股份)	4	1,000	4,000	1.9%	4	600	2,400	2.5%	
	Joint torque sensor	Keli Sensing Technolgn (柯力传感), Donghua Testing (东华测试), Kunwei Technology (坤维科技), Sunrise (宇立仪器)	28	750	21,000	10.2%	28	300	8,400	8.7%	
	Six-dimensional torque sensor		4	2,000	8,000	3.9%	4	1,000	4,000	4.1%	
	Tactile Sensor	Hanwei Electronics (汉威科技), Shenhao Technology (申昊科技)	10	200	2,000	1.0%	10	100	1,000	1.0%	
		Subtotal of Tactile Sensor			36,500	17.8%		16,700	17.3%		

Exhibit 21: Humanoid robots' bill of materials (BOM) cost estimates (RMB)

Control system	Controller	STEP Electric (新时达), Topband (拓邦股份), Leadshine Technology (雷赛智能), Invt Electric (英威腾), Maxonic Automation(万讯自控)	40	300	12,000	5.8%	40	150	6,000	6.2%
Software	AI chips & software	Nvidia, Tesla, Figure AI	1	19,000	19,000	9.2%	1	10,000	10,000	10.3%
		Others			75,00	3.6%			2,500	2.6%
Total Cost					205,520	100%			96,720	100%

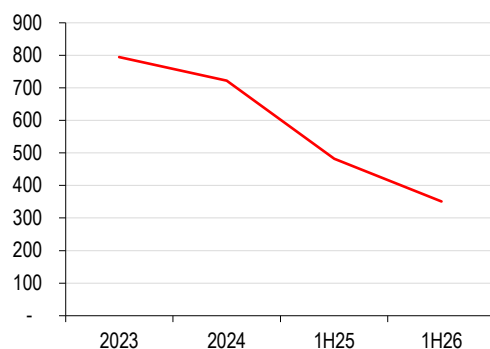
Source: Company data, HSBC estimates

Exhibit 22: Industrial robot and EV historical cost reduction



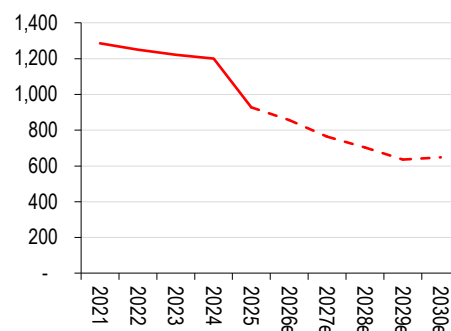
Source: Tesla, HSBC estimates

Exhibit 23: Laifual Drive (3952 HK) harmonic reducer ASP (RMB)



Source: Company data, HSBC

Exhibit 24: Harmonic reducer ASP in China (RMB)



Source: CIC, HSBC

Exhibit 25: Figure AI's humanoid robot component suppliers

Company	Chinese name	Ticker	Components supplied
Xusheng Group	旭升集团	603305 CH	Magnesium alloy shell
Everwin Precision	长盈精密	300115 CH	Joint, bearing, sensor
Zhaowei Machinery & Electronics	兆威机电	003021 CH	Coreless motor
Leaderdrive	绿的谐波	688017 CH	Harmonic drive
Lingyi iTECH	领益智造	002600 CH	Structure, thermal module
MOONS	鸣志电器	603728 CH	Coreless motor, servo motor
Yinlun Machinery	银轮股份	002126 CH	Rotate vector

Source: HSBC compilation

Exhibit 26: Unitree's humanoid robot component suppliers

Company	Chinese name	Ticker	Components supplied
MOONS	鸣志电器	603728 CH	Coreless motor
Zhaowei Machinery & Electronics	兆威电机	003021 CH	Coreless motor
Zhongda Leader	中大力德	002896 CH	Planetary reducer
Luster	凌云光	688400 CH	Vision solution
Wanma	万马股份	002276 CH	Wire
Wolong	卧龙电驱	600580 CH	Servo motor, control system
Leaderdrive	绿的谐波	688017 CH	Harmonic drive reducer
CSB Bearing	长盛轴承	300718 CH	Bearing, ball screw
Beite	北特科技	603009 CH	Balls crew
Allwinner Technology	全志科技	300458 CH	Chip
Orbtec	奥比中光	688322 CH	Vision solution, sensor, radar
Leili	江苏雷利	300660 CH	Motor, actuator
Wuxi Best	贝斯特	300580 CH	Ball screw, linear drive
Kinfa	金发科技	600143 CH	Engineering plastics
Shuanglin	双林股份	300100 CH	Ball screw

Source: HSBC compilation

Exhibit 27: Tesla's humanoid robot component suppliers

Company	Chinese name	Ticker	Components supplied
Sanhua	三花智控	002050 CH	Actuator assembly, Tier 1 supplier
Tuopu	拓普集团	601689 CH	Actuator assembly, Tier 1 supplier
Hengli	恒立液压	601100 CH	Roller screw/ Ball screw
Beite	北特科技	603009 CH	Ball screw
XCC	五洲新春	603667 CH	Ball screw
Best	贝斯特	300580 CH	Ball screw
Sling	斯菱股份	301550 CH	Ball screw, Reducer
Shuanglin	双林股份	300100 CH	Ball screw
Leili	江苏雷利	300660 CH	Ball screw, Tier 2 supplier
Jw	金沃股份	300984 CH	Ball screw, Tier 2 supplier
Leader Harmonious	绿的谐波	688017 CH	Reducer
Shuanghuan	双环传动	002472 CH	Reducer
Zhaowei	兆威机电	003021 CH	Dexterous hand
Moons'	鸣志电器	603728 CH	Dexterous hand, Tier 1 supplier
Hajime	肇民科技	301000 CH	Material
Junchuang	骏创科技	833533 CH	Face mask
Ampron	安培龙	301413 CH	Temperature and pressure Sensor
Fore	丰立智能	301368 CH	Reducer
Xusheng	旭升集团	603305 CH	Magnesium alloy casing
Rongtai	嵘泰股份	605133 CH	Light-weight material

Source: HSBC compilation

How to position: Previously, we believe hardware component players (balls crew and reducers) would benefit first from humanoid robot mass production, however, we observe increasing price competition, given many players are expanding capacity before mass shipment arrives. Medium-term shipments will depend more on a stronger 'brain' that leads to mass deployment in factories, which is still limited now.

Key humanoid technical specs

Exhibit 28: Summary of humanoid robots, mainland China

Company	UBTech	UniTree	Galbot	Galaxea	Deep Robotics	Dobot	Agibot	X Square	Robbyant
Chinese name	优必选	宇树	银河通用	星海图	云深处	越疆	智元	自变量	蚂蚁灵波
Listed/unlisted	9880 HK	688836 CH	Private	Private	Private	2432 HK	Private	Private	Private
Humanoid	Walker S2	G1/G1 Edu	ET1	R1 Pro	DR1/DR2	Atom	A3	Quanta X2	R2
Humanoid launch	2025	2024	2026	2026	2025	2025	2026	2025	Coming soon
Technical specs									
Height (cm)	176	130	125	170	175	153	173	164	na
Weight (kg)	70	35	32	126 (wheeled)	75	62	55	na	na
Payload (kg)	15	na	na	10	10	na	10	12	na
Speed (km/h)	7.2	7.2	na	36	20	na	na	7.2	na
Sales price (RMBk)	NA	99	na	70	na	na	na	na	na
DoF	52	23/43	26/26+12	26	31	41	51	62	na

Source: Company websites, HSBC

Exhibit 29: Summary of humanoid robots, overseas

Company	Tesla	Physical Intelligence	Generalist AI	Skill AI	Field AI	Figure AI	Agility Robotics	1X
Listed/Unlisted	TSLA US	Private	Private	Private	Private	Private	AGLT US	Private
Humanoid	Optimus Gen3	Humanoid: Pi Brain focus (latest model: $\pi 0.7$)	Brain focus (latest model: S1)	Brain focus (latest model: GEN-1.5)	Robot dog focus	Figure 03	Digit	Neo
Humanoid launch	2026	2024				2025	2019	2025
Technical specs								
Height (cm)	180					170	175	168
Weight (kg)	45					61	37	30
Payload (kg)	na					20	18	25
Speed (km/h)	20					4.3	na	5
Price (USDk)	49					NA	250	20
DoF	75					NA	16	75

Source: Company websites, HSBC

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